

Lab Programs

SET-1

1. Write a Pandas program to select the distinct department id from employees file.

Aim

To write a Pandas program to display the distinct department IDs from the employees dataset.

Algorithm

1. Import the Pandas library.
2. Read the employees CSV file.
3. Select the **department_id** column.
4. Use the unique() function to find distinct department IDs.
5. Display the result.

Dataset (employees.csv)

Employee_ID	First_Name	Last_Name	Department_ID	Job_Title	Salary
101	John	Smith	10	Manager	75000
102	Alice	Brown	20	Analyst	55000
103	David	Wilson	10	Manager	72000
104	Emma	Thomas	30	Clerk	35000
105	James	Martin	20	Analyst	56000

Employee_ID	First_Name	Last_Name	Department_ID	Job_Title	Salary
106	Sophia	Lee	40	Developer	65000

Code

```
import pandas as pd
df = pd.read_csv(r"C:\Users\HP\Downloads\dataset.txt")
print("Distinct Department IDs:")
print(df["department_id"].unique())
```

Output

```
Distinct Department IDs:
[10 20 30 40]
```

Result

Thus, the distinct department IDs were displayed successfully.

2. Pandas Program to Display the IDs of Employees Who Have Done Two or More Jobs

Aim

To display employee IDs of employees who have worked in two or more jobs.

Algorithm

1. Import Pandas.
2. Read the employee history dataset.

3. Group records by employee ID.
4. Count the number of jobs.
5. Display employee IDs having job count greater than or equal to 2.

Dataset (job_history.csv)

Employee_ID	Job_Title
101	Manager
101	Analyst
102	Developer
103	Clerk
103	Manager
104	Analyst
105	Developer
106	Clerk

Code

```
1 import pandas as pd
2 df = pd.read_csv(r"C:\Users\HP\Downloads\dataset.txt")
3 job_count = df.groupby("employee_id").size()
4 print(job_count[job_count >= 2])
```

Output

```
employee_id
101      2
103      2
dtype: int64
```

```
employee_id
```

```
101    2
```

```
103    2
```

```
dtype: int64
```

Result

Thus, the employee IDs having two or more job records were displayed successfully.